



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 • STANDARD 6.NOS.4

# Prime Factorization

Lesson 1-1 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can write a number as a product of its prime factors using a factor tree.

**Language Objective:** I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Prime number

Composite number

Prime factorization

Factor

Factor tree

Exponent



Tap any word to see what it means and a picture.

1

A whole number greater than 1 with exactly two factors, 1 and itself, is a  number.

2

A whole number greater than 1 that has more than two factors is a  number.

3

— Writing a number as prime numbers multiplied together.

4

A number that divides evenly into another number is called a .

5 I can break a number into its prime factors step by step using a

6 In  $2^3$ , the small number 3 is the  and it shows 2 is multiplied 3 times.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Look at your factor tree for 60. How did you decide which numbers to break apart first, and how do you know when to stop?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Prime number** — A number bigger than 1 that you can only divide by 1 and itself.  
*Número primo — Un número mayor que 1 que solo se puede dividir entre 1 y sí mismo.*

- ☐ **Composite number** — A number bigger than 1 that you can divide by more than just 1 and itself.  
*Número compuesto — Un número mayor que 1 que se puede dividir entre más números, no solo 1 y sí mismo.*

- ☐ **Prime factorization** — Writing a number as prime numbers multiplied together.  
*Factorización prima — Escribir un número como números primos multiplicados.*

- ☐ **Factor** — A number that divides evenly into another number, with no remainder.  
*Factor — Un número que divide exactamente a otro número, sin dejar residuo.*

- ☐ **Factor tree** — A picture that splits a number into its prime numbers, step by step.  
*Árbol de factores — Un dibujo que separa un número en sus números primos, paso a paso.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**I broke \_\_\_\_ into \_\_\_\_ because \_\_\_\_.**

*Separé \_\_\_\_ en \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** 60 is composite because it has more than two factors.

**Evidence:** Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

**Reasoning:** This evidence connects the definition of prime factorization to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **I broke \_\_\_\_ into \_\_\_\_ because \_\_\_\_.**  
*Separé \_\_\_\_ en \_\_\_\_ porque \_\_\_\_.*
- **I knew I was finished when every factor was \_\_\_\_.**  
*Supe que terminé cuando cada factor era \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **I know because \_\_\_\_.**  
*Lo sé porque \_\_\_\_.*
- **So \_\_\_\_.**  
*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor
5. **Notes 5:** Factor tree
6. **Notes 6:** Exponent
7. **Watch for this mistake:** *A common mistake in Prime Factorization is stopping the factor tree too soon — leaving a composite number like 4, 6, 8, or 9 in the final answer instead of continuing to split it. For example, writing  $20 = 4 \times 5$  as the prime factorization (4 is composite) instead of finishing with  $20 = 2 \times 2 \times 5$ . Before you submit, check every number in your final answer: if any factor can still be broken into smaller factors, the tree isn't finished yet.*
8. **Try It 1:** A.  $36 - 2^2 \times 3^2 = 4 \times 9 = 36$ , so 36 is the code for this panel.
9. **Try It 2:** A.  $2 \times 3 \times 13 - 78 = 2 \times 39 = 2 \times 3 \times 13$ . All three factors are prime, so  $2 \times 3 \times 13$  unlocks the final panel.